

STATS 418

Color Analysis using Machine Learning

Simplifying style and building a cohesive closet

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Overview/Motive

Why color analysis?

Problem: While some enjoy the process, getting dressed is unnecessarily difficult for many from celebrities to everyday people.

Motivation: Your clothes should fit you rather than you needing to fit your clothes. Using color analysis, people can filter which clothes suit them and which do not to optimize the process.



Data Collection Approach



Data Source:

ColorMine.ai – Celebrity color analysis directory



Method:

Web scrape each celebrity's information

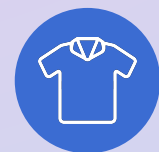


Data:

Collected each celebrity's job, job category, eye color, hair color, skin tone, contrast, **color season**, and best/worst colors, and image

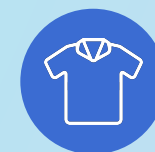
Model Dev/Eval

Random Forest vs. CNN



Random Forest:

- 320 observations in a 70/15/15 split
- **Gridsearch:**
 - estimators: 100
 - max depth: 10
 - split: 5
 - leaf: 2
 - max features: sqrt
 - **class weight: balanced**
- **Test Performance:**
 - F1 Score: 0.3618
 - Accuracy: 45.83%

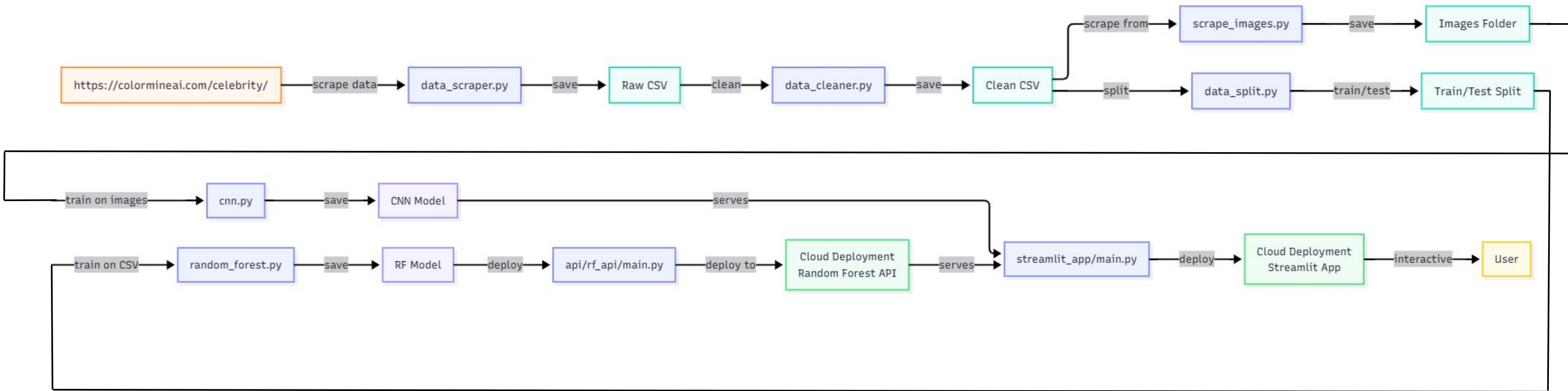


CNN:

- 320 observations in a 85/15 split
- **Architecture:**
 - Four blocks
 - Padding: 1
 - Stride: 2
 - Kernel: 3
- **Test Performance:**
 - F1 Score: 0.01
 - Accuracy: 14.58%
 - Loss: 2.729



Architecture Diagram



Application



main
manual
picture

Color Analysis Prediction App

This app predicts a person's color analysis as one of 12 categories:

- Winter: clear, cool, deep
- Spring: clear, light, warm
- Summer: cool, light, soft
- Autumn: deep, soft, warm.

You can get your color analysis two ways:

1. Manually input your characteristics such as eye and hair color and get results

OR

2. Input a picture and get your color analysis.

Use the sidebar to navigate between the two options.

Your features

Choose your contrast level

☒ medium
☐ high
☐ low

Choose your eye color

brown

Choose your hair color

dark brown

Choose your skin tone

☒ cool
☐ warm

Predict Season

Manual Color Analysis

This allows you to input your contrast, eye color, hair color, and skin tone to predict your color season. Enter your details below to get your color analysis prediction.

Your details

Feature	Value
Contrast	medium
Eye Color	brown
Hair Color	dark brown
Skin Tone	cool

> About the Model

> High, Medium, or Low contrast

> Warm or Cool

Color Season

Color Analysis

cool summer

main
manual
picture

Upload your image

Upload a .png image of you for analysis

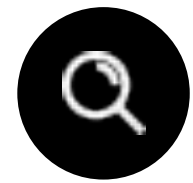
prof418.png
292.6KB



Color Season

Predict

Predicted class: deep autumn



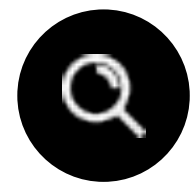
FastAPI: CNN

- More than 520 Mib limit
- Kept local, but load time is SLOW



Color

- Too many unique colors for my model scale
- Took most frequent colors in each season to recommend

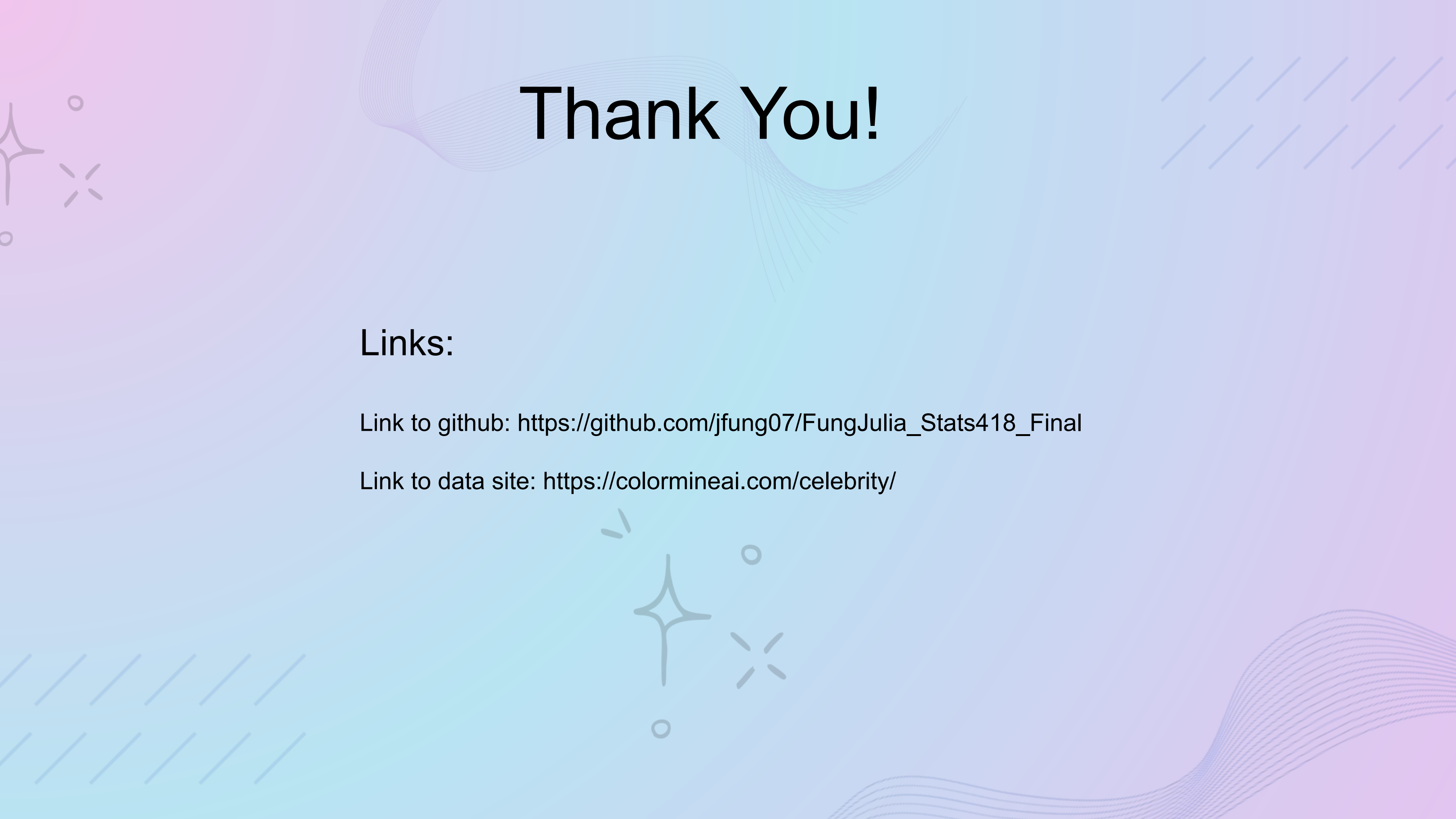


Future Work

- Birthday
- Color Recommendations

Problems and Solutions





Thank You!

Links:

Link to github: https://github.com/jfung07/FungJulia_Stats418_Final

Link to data site: <https://colormineai.com/celebrity/>